

Lab Record: Dijkstra's Algorithm Using Min Heap

Aim

To design and implement a C program to find the shortest paths from a given source vertex to all other vertices in a weighted connected graph using Dijkstra's Algorithm with Min Heap.

Theory

Dijkstra's Algorithm is used to find the shortest path between a source vertex and all other vertices in a weighted graph with non-negative edge weights. A Min Heap is used to efficiently select the vertex with minimum distance.

Algorithm

1. Take Min Heap node containing two values (v, w).
2. Take distance array of size v and initialize all values as infinity.
3. Take parent array of size v.
4. Read source vertex (sv).
5. Change parent of sv as sv.
6. Change distance from sv to sv as zero.
7. Insert (sv, 0) into Min Heap. (v, w) where Min Heap is based on w.
8. Loop until heap becomes empty:
 - a. Perform delete operation on Min Heap.
 - b. Based on deleted element move to adjacency matrix of u/x and find non-zero v/y.
 - c. Apply formula: $\text{dis}[v] = \min(\text{dis}[u] + w, \text{dis}[v])$
 - d. If the formula works, update the distance of v.
 - e. Change the parent of v and insert into heap.

Time Complexity

Best Case: $O(V^2)$

Average Case: $O(V^2)$

Worst Case: $O(V^2)$

Applications

- Routing protocols

- GPS navigation systems
- Network analysis
- Shortest path calculations

Conclusion

Dijkstra's Algorithm efficiently computes the shortest path from a source vertex to all other vertices using greedy strategy and Min Heap concept.